Chapter 6

Automated & Emerging Technologies

Cambridge IGCSE & O Level Computer Science

(2210 / 0478)

Covers: 6.1 Automated systems - 6.2 Robotics

6.3 Artificial Intelligence (AI)

Ready-to-learn notes with definitions, diagrams, exam tips & past-paper style practice

6.1 Automated Systems

6.1.1 The Sensor - Microprocessor - Actuator Loop

Automated system: hardware + software designed to work automatically without human intervention (though it's often still human-monitored).

SENSOR reads the environment (analogue -> ADC -> digital)

↓

MICROPROCESSOR processes data, decides what action is needed

↓

ACTUATOR carries out the physical action (motor, valve, solenoid...)

This 3-step loop is the backbone of EVERY automated system example

Exam tip – for any automated-system scenario question, always identify: which sensor(s)? what does the microprocessor decide? what does the actuator physically do?

6.1.2 Example: Nuclear Power Station

Sensors: temperature, pressure, flow level, gas, radiation. Actuators: water pumps, valves, gas pumps, automatic shutdown. A Distributed Control System (DCS) monitors everything; a human supervisor can override it.

Advantages	Disadvantages
Faster than a human operator	Very expensive to set up + test
Safer – keeps humans out of danger	Untested conditions could still occur
Runs at optimum conditions constantly	Vulnerable to cyberattacks
Cheaper long-term (less workforce needed)	Needs expensive, enhanced maintenance

Exam tip – this adv/disadv pattern (speed+safety+cost vs setup-cost+cyber-risk+maintenance) repeats across almost every automated-system example – learn the PATTERN, not just this one case.

6.2 Robotics

6.2.1 What is a Robot?

> *Robotics:* the branch of computer science covering the design, construction and operation of robots.

Found in: factories (welding, spray-painting, precision cutting), the home (floor sweepers, lawn mowers, window cleaners), and drones (UAVs – reconnaissance, deliveries).

6.2.2 The 3 Characteristics of a TRUE Robot

1. Can sense its surroundings – via sensors (light, pressure, temperature, acoustic...)
2. Has a degree of movement – wheels, cogs, pistons, gears, end-effectors (welding tip, spray gun, gripper)
3. Is programmable – has a "controller" (its brain) that decides actions based on sensor data

> *Exam trap* – a "robot" needs ALL 3. A software bot (web crawler, chatbot) is NOT a true robot by this definition – and most robots do NOT possess AI (they just repeat programmed tasks)!

Independent vs Dependent Robots

Independent (autonomous)	Dependent
No direct human control	Human interfaces directly with it (control panel/computer)
Totally replaces the human task	Supplements human activity (works alongside humans)
e.g. self-driving car	e.g. robot arm in a car assembly line

6.2.3 Robots - Advantages & Disadvantages

Advantages	Disadvantages
Work in hazardous conditions humans can't	Struggle with "non-standard" / unexpected tasks
Work 24/7 without stopping	Causes unemployment in manual labour
Cheaper long-term (fewer salaries)	Risk of "deskilling" the workforce
More productive & consistent than humans	Expensive to buy and set up initially
Better suited to repetitive/boring tasks	Factories can relocate for cheaper labour elsewhere

Autonomous Vehicles - Worked Example

Cameras + radar/ultrasonic sensors act as "eyes and ears". At a red light: microprocessor recognises the sign, checks its database, sends signals to actuators -> apply brakes + shift to park. On green: release brakes, engage throttle - constantly re-checking all sensors for safety throughout.

> *Exam tip* - "describe the role of sensors/microprocessor/actuators" scenario questions always want this 3-part structure: what data is sensed -> what decision the microprocessor makes -> what physical action the actuator performs.

6.3 Artificial Intelligence (AI)

> *AI:* a branch of computer science dealing with the simulation of intelligent human behaviour (reasoning, learning, adapting) by a computer.

6.3.1 Three Categories of AI

Narrow AI	General AI	Strong AI
Superior to a human at ONE specific task	Similar (not superior) to a human at a specific task	Superior to a human across MANY tasks

Examples of AI

Smart home devices (Alexa, Siri) – chatbots – autonomous cars – facial expression recognition (maps landmarks like eyebrow/mouth corners to emotions) – news generation from live feeds.

6.3.2 Two Types of AI System

> *Expert system:* mimics the decision-making of a human expert using stored knowledge and rules.

> *Machine learning:* trains algorithms on sample data so they can predict outcomes for NEW, unseen data – without being explicitly re-programmed.

Expert Systems - Structure

User Interface <-> Inference Engine <-> Knowledge Base + Rules Base

Fig 6.23 - typical expert system shell

- User interface: asks the user Yes/No style questions (dialogue boxes, prompts)
- Inference engine: the "brain" - searches the knowledge base for matches based on user answers
- Knowledge base: repository of facts - a collection of objects and their attributes

Used for: medical diagnosis, oil/mineral prospecting, fault diagnostics, tax calculations, chess strategy, logistics routing.

Advantages	Disadvantages
High expertise, accuracy & consistency	Users need training to use it properly
Very fast response vs a human expert	Very expensive to set up and maintain
Gives probability of its conclusion	Responses can feel "cold" (bad for sensitive contexts)
Unbiased, traceable logic	Only as good as the data entered into it

Machine Learning - Worked Examples

> *Exam tip* - AI = "building machines that think like humans." Machine learning = "getting machines to make decisions from data, without being explicitly re-programmed." Machine learning is a SUB-SET of AI - don't treat them as separate/equal.

Example: Spam Email Detection

1. Collect data (content, headers, sender)
2. Clean data (remove stop words, punctuation)
3. Spot patterns - words like "lottery", "earn", "full-refund" flag spam
4. Train a model on a training data set of known spam
5. Evaluate, fine-tune, then test live

Example: Buying History (Collaborative Filtering)

Compares a new customer's habits against similar past customers to recommend products - e.g. two customers who both bought a jazz CD + a Roman history book get cross-recommended each other's other purchases.

Example: Fraud Detection

Data collection -> Data cleaning -> Exploration/analysis -> Build a model -> Model evaluation

Fig 6.27 - general machine learning model pipeline (applies to ALL ML examples)

Past-Paper Style Practice

Cambridge-style questions written for self-practice, with a model marking scheme below each.

Q1

[3]

State the three characteristics needed for a machine to be classified as a true robot.

Mark scheme (1 mark each, max 3):

- Ability to sense its surroundings (via sensors)
- Has a degree of movement
- Is programmable (has a controller)

Q2

[4]

Describe, using sensors, a microprocessor and actuators, how an automated system might control a greenhouse's temperature.

Mark scheme (max 4):

- Temperature sensor reads the current temperature (converted via ADC if analogue)
- Data sent to the microprocessor, which compares it to a stored/desired value
- If temperature is too high/low, the microprocessor sends a signal to an actuator
- Actuator opens/closes a vent or turns a heater on/off

Q3

[2]

Explain the difference between narrow AI and strong AI.

Mark scheme (max 2):

- Narrow AI outperforms a human at one specific task only
- Strong AI outperforms a human across many different tasks

Q4

[3]

Describe the role of the inference engine in an expert system.

Mark scheme (max 3):

- It is the main processing/problem-solving element of the expert system
- It searches the knowledge base for information matching the user's responses
- It applies rules from the rules base to find a matching object/conclusion

Q5

[3]

Give three disadvantages of using robots in industry.

Mark scheme (1 mark each, max 3):

- Struggle to handle non-standard/unexpected tasks
- Can lead to unemployment in manual labour
- Expensive to buy and set up initially / risk of deskilling workers

Before your exam – Practice re-drawing the sensor-microprocessor-actuator loop and applying it to a NEW scenario you haven't seen before (e.g. a car park barrier, a washing machine). That's exactly how these questions are asked.